

MAY HTET HTET KHINE

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[Linkedin](#) [Portfolio](#)

SUMMARY

Data and machine learning professional with experience in manufacturing analytics, Bayesian modeling, and foundation model applications in industrial environments. Proven ability to develop data-driven solutions for failure analysis and process optimization, including achieving measurable reliability improvements. Strong background in Python, statistical modeling, and large-scale data processing, with a focus on translating complex data into actionable business insights.

EXPERIENCE

Data Science Researcher — Seagate Technology (Thailand) Aug 2025 - Present

- Developed data-driven optimization frameworks for HDD manufacturing processes, integrating Design of Experiments (DoE), statistical modeling, and process analysis to reduce variability and improve yield
- Applied foundation models (e.g., TabPFN and time-series approaches) to predict target variables from high-dimensional wafer sensor data, particularly in low-data regimes
- Benchmarked model performance against traditional approaches (e.g., XGBoost, Random Forest), analyzing generalization and robustness across varying data conditions
- Performed feature engineering, exploratory data analysis (EDA), and feature importance analysis to identify key process drivers
- Implemented Bayesian and probabilistic models to support uncertainty-aware decision-making under manufacturing variability
- Collaborated with cross-functional engineering teams to translate analytical insights into actionable process improvements

Data Science & AI Intern — Seagate Technology (Thailand) Jan 2025 - Apr 2025

- Built Bayesian models (PyMC) to identify root causes of drive failures and rank critical parameter configurations
- Improved diagnostic reliability by 17% (IoU metric) through probabilistic modeling
- Designed decision-support tools for long-term infrastructure integration
- Performed data preprocessing, feature engineering, and statistical analysis on engineering datasets

Research Assistant — Mae Fah Luang University (Thailand) Jun 2023 - Dec 2024

- Developed machine learning and Bayesian models for nonlinear state estimation and prognostics
- Conducted time-series and sequential data analysis for engineering applications
- Co-authored peer-reviewed publications and collaborated on research methodology development

SKILLS

Technical Skills

- Programming: Python (Pandas, NumPy, Scikit-learn, PyMC), SQL, ApacheSpark (PySpark)
- Machine Learning: Regression, Classification, Bayesian Modeling, Time Series, Foundation Models
- Data & Analytics: Data Cleaning, Feature Engineering, Exploratory Data Analysis (EDA), Statistical Analysis
- Tools: n8n, Power BI, Microsoft Excel, DBeaver, MATLAB, AI Agents
- Methods: Design of Experiments (DoE), Probabilistic Modeling, ETL/ELT Pipeling, Agile-based SDLC

Languages

- Burmese (Native)
- English (TOEIC: 965)
- Thai (Conversational)

PUBLICATIONS

- Khine, M. H. H., Kim, C. G., Aunsri, N. A review of Bayesian-filtering-based techniques in RUL prediction for Lithium-Ion batteries, Journal of Energy Storage, 2025.
- Khine, M. H. H., Aunsri, N. Particle Weight Redistribution in Particle Filtering to Improve Nonlinear State Estimation, IEEE InCIT, 2024.
- Khine, M. H. H., Kuptamettee, C., Aunsri, N. Improving Reliability of State Estimation by Particle Filter with Cauchy Likelihood Function, IEEE InCIT, 2023.

EDUCATION

B.Eng. (First Hons.) Computer Engineering — Mae Fah Luang University Aug 2021 - May 2025

- GPAX: 4.00
- Outstanding Academic Achievement Medal
- Concentration: Data Science and Artificial Intelligence

Diploma in Computing (Level 5, Ofqual England) Jan 2020 - Jan 2022

- Foundation in programming, database systems, and information systems
- Developed early experience in data handling, web development, and system design